PAC-Bayesian Treatment Allocation Under Budget Constraints

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Motivating Example

- Observe randomized assignment of malaria prevention "treatment" (example: mosquito nets provided to household). Outcome of interest recorded some time after treatment.
- Want individualized treatment rule –given an individual/household characteristics, it determines if treatment is given– that maximizes the outcome of interest in aggregate when applied to all individuals in the population where data is from.
- Constraint: aid organization or govt may not have resources to provide treatment to all. Must maximize outcome given resources available.

Motivating Example

- Uncertainty and heterogeneity in benefits and costs. Benefit of treatment could vary with number of children in a household. Cost could vary with type of distribution channel nearest to household.
- [Bhattacharya and Dupas, 2012]: Estimate treatment rules that satisfy quantity constraints with fixed costs for mosquito net assignments. Rules considered here can target policies that account for cost heterogeneity to improve rule efficiency and target higher outcome for same cost.
- Problem also arises in marketing and medical applications.

Contributions

- Expands limited tool set for estimating individual treatment rules when there is a general budget constraint.
- Rules can be applied in settings where alternatives can't.
- Ability to incorporate expert knowledge to improve performance.
- Derive theoretical properties (budget efficiency) that alternative methods do not show.
- To the PAC-Bayesian literature:
  - PAC-Bayesian learning approach that accounts for secondary objective of interest. Arguments can be generalized to settings like classification.
  - Budget-constrained treatment setting considered in PAC-Bayesian framework for first time.

Outline From Here

Problem of Interest and Statistical Setting

Literature Review

Pac-Bayesian Setting

Motivating the Gibbs Rules

Theoretical Properties

Simulation Study

Empirical Illustration

End

Problem of Interest and Statistical Setting

Problem of interest

$$Y = Y_1 D + Y_0(1-D), \;\; C = C_1 D + C_0(1-D).$$

- Only observe: $D \in \{0,1\}$, outcome Y, cost C, characteristics $X \in \mathcal{X} \subseteq \mathbb{R}^{d_x}$.
- ITR denoted $f : \mathcal{X} \to [0,1]$. If $f(x) \in \{0,1\}$, non-stochastic at x.
- **Utilitarian welfare** of f: $E[Y_1 f(X) + Y_0(1-f(X))]$. Define

$$W(f) \equiv E[(Y_1 - Y_0)f(X)]; \;\; K(f) \equiv E[(C_1 - C_0)f(X)].$$

- Policy maker's problem with max budget B:

$$\max_f W(f) \;\; \text{s.t.} \;\; K(f) \leq B.$$

Statistical Setting: Model family

$$\mathcal{F}_\Theta = \{f_\theta(x) : \mathcal{X} \to \{0,1\};\ \theta \in \Theta\}$$

is a family of non-stochastic treatment rules indexed by parameter vector $\theta \in \Theta$. Paper derives some results under more general settings that could allow for infinite dimensional Θ.

Here, take

$$f_\theta(x) = 1\{\phi(x)'\theta > 0\},\ \theta \in \Theta = \mathbb{R}^q,$$

where $\phi_j(x) : \mathcal{X} \to \mathbb{R},\ j = 1, \ldots, q$.

Throughout, all probability measures over the model/parameter space are over $(\mathbb{R}^q, \mathcal{B}(\mathbb{R}^q))$ and integrals over this space are denoted $\int$ rather than $\int_{\mathbb{R}^q}$

Statistical Setting: Assumption (assumed throughout)

(i) Sample: n iid observations of $Z = (Y, C, D, X)$, where

$$Y = Y_1 D + Y_0(1 - D), C = C_1 D + C_0(1 - D).$$

Will denote $Z_i = (Y_i, C_i, D_i, X_i)$ for $i = 1, \ldots, n$. Assumed the joint distribution over (Y_0, Y_1, C_0, C_1, X) in the population where the treatment rule will be applied is the same as that generating the sample.

(ii) Unconfoundedness: $(Y_1, Y_0, C_1, C_0) \perp D | X$.

(iii) Bounded Outcomes/Costs: $\exists\ 0 < M_y, M_c < \infty$ such that: support of Y contained in $[-\frac{M_y}{2}, \frac{M_y}{2}]$, support of C contained in $[-\frac{M_c}{2}, \frac{M_c}{2}]$.

(iv) Strict Overlap: $\exists\ \kappa \in (0, 1/2)$ such that $e(x) \in [\kappa, 1 - \kappa]$ for all $x \in \mathcal{X}$ where

$$e(X) = E[D|X].$$

Statistical Setting: Definitions, Objects of Interest

$$W(\theta) \equiv W(f_\theta) = E[(Y_1 - Y_0) f_\theta(X)]$$
$$K(\theta) \equiv K(f_\theta) = E[(C_1 - C_0) f_\theta(X)]$$

Let

$$\delta_{y,i} = \left(\frac{Y_i D_i}{e(X_i)} - \frac{Y_i(1 - D_i)}{1 - e(X_i)} \right), \delta_{c,i} = \left(\frac{C_i D_i}{e(X_i)} - \frac{C_i(1 - D_i)}{1 - e(X_i)} \right).$$

Empirical counterparts:

$$W_n(f) = \frac{1}{n} \sum_{i=1}^{n} \delta_{y,i} f(X_i); \qquad W_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} \delta_{y,i} f_\theta(X_i).$$

$K_n(f)$, $K_n(\theta)$ defined similarly. $K_n(\theta)$, $W_n(\theta)$ unbiased.

Literature Review

Literature Review – Treatment Personalization, incomplete list

Feature: only observe an individual's outcome associated with treatment allocation given, never counterfactual.

- Econometrics: [Kitagawa and Tetenov, 2018]: empirical welfare maximization (EWM). [Mbakop and Tabord-Meehan, 2021]: model selection/penalized EWM. [Athey and Wager, 2021], ... These examples allow for general functional form constraints.
- Biostatistics: [Qian and Murphy, 2011]: plug in approach, [Zhao et al., 2012]: SVM for treatment rules.
- Computer science & machine learning: [Beygelzimer and Langford, 2009]: adapt binary classification, regret bounds. Logged Bandit problems: [London and Sandler, 2019] consider PAC-Bayesian counterfactual risk minimization.
- [Kitagawa et al., 2023] consider stochastic treatment assignment rules from a PAC-Bayesian approach, but do not consider the setting with a general budget constraint that is the focus here.

Literature Review – Treatment Rules with resource constraints

- [Sun, 2021]: extends arbitrary form features of (EWM) to budget constrained setting. Can be difficult to estimate for more flexible functional forms, rules that approach welfare of optimal policy allowed to violate budget. Allows cost savings.
- [Sun et al., 2021]: when treatment can be assigned in batches (observe costs relatively quickly) and cost of treatment always exceeds that of non-treatment, derive plug in approach utilizing the generalized random forest method of [Athey et al., 2019].
- Outcome weighted treatment learning utilized in [Wang et al., 2018], [Huang and Xu, 2020]. Somewhat ad hoc. Start from empirical version of policy maker's problem and or plug-in. Properties of proposed rules are not analyzed.

Pac-Bayesian Setting

PAC-Bayesian Learning

- Features of PAC-Bayesian Analysis:

  a. Treatment assignment rules, classifiers, etc. that feature or can be related to **randomization or aggregation over a family of rules or models**. We will start with a non-data-dependent reference distribution or "prior" that is re-weighted by the sample to obtain an evidence-based distribution, the "posterior," that is utilized by the treatment rule.
  b. Probabilistic **generalization bounds** for properties of these algorithms. Probably Approximately Correct framework.

PAC-Bayesian Setting: Gibbs treatment rules

1. **Gibbs treatment rule for** ρ over $(\mathbb{R}^q, \mathcal{B}(\mathbb{R}^q))$. Given covariates X,

$$\text{draw } \theta_\circ \sim \rho, \text{ then treat via } f_{\theta_\circ}(X) \in \{0, 1\}.$$

Independent draw from ρ each time treatment is assigned. Here,

$$f_{G,\rho}(x) = \int f_\theta(x) d\rho(\theta), \ x \in \mathcal{X}.$$

2. The (non-stochastic) **majority vote rule for** ρ is

$$f_{\mathrm{mv},\rho}(x) = 1\left\{\int f_\theta(x) d\rho(\theta) > \frac{1}{2}\right\}, \ x \in \mathcal{X}.$$

$f_{\mathrm{mv},\rho}$ inherits some desirable properties from $f_{G,\rho}$ when $f_{G,\rho}$ produces relatively high welfare for its cost. Focus on properties of $f_{G,\rho}$ here.

PAC-Bayesian Setting: additional notation

$$W(f_{G,\rho}) = E\left[(Y_1 - Y_0)\int f_\theta(X)d\rho(\theta)\right] = \int W(\theta)d\rho(\theta)$$

$$W_n(f_{G,\rho}) = \frac{1}{n}\sum_{i=1}^{n}\left[\delta_{y,i}\int f_\theta(X)d\rho(\theta)\right] = \int W_n(\theta)d\rho(\theta)$$

Special notation for cost or budget of Gibbs rule using ρ:

$$B(\rho) \equiv K(f_{G,\rho}) = E\left[(C_1 - C_0)\int f_\theta(X)d\rho(\theta)\right] = \int K(\theta)d\rho(\theta)$$

$$B_n(\rho) \equiv K_n(f_{G,\rho}) = \frac{1}{n}\sum_{i=1}^{n}\left[\delta_{c,i}\int f_\theta(X)d\rho(\theta)\right] = \int K_n(\theta)d\rho(\theta)$$

For $W(f_{G,\rho})$, $K(f_{G,\rho})$, E taken with respect to a draw from (Y_0, Y_1, C_0, C_1, X) that is independent of ρ if ρ is sample-dependent.

PAC-Bayesian Setting: proposed family of posteriors

Assumption: π is a (determinisitc) probability measure over $(\mathbb{R}^q, \mathcal{B}(\mathbb{R}^q))$ that doesn't depend on the sample.

We will utilize $f_{G,\hat{\rho}_{\lambda,u}}$ and $f_{\mathrm{mv},\hat{\rho}_{\lambda,u}}$, where

Gibbs Posterior

Let $\lambda > 0$, $u \geq 0$. Define $\hat{\rho}_{\lambda,u}$ as the (random) probability measure on $\mathbb{R}^q$ with the following RN-derivative w.r.t. π:

$$\frac{d\hat{\rho}_{\lambda,u}}{d\pi}(\theta) = \frac{\exp\left[-\lambda\left(uK_n(\theta) - W_n(\theta)\right)\right]}{\int \exp\left[-\lambda\left(uK_n\left(\tilde{\theta}\right) - W_n\left(\tilde{\theta}\right)\right)\right] d\pi\left(\tilde{\theta}\right)}.$$

PAC-Bayesian Setting: additional details

Empirical budget,

$$B_n(\hat{\rho}_{\lambda,u}) = \int K_n(\theta) d\hat{\rho}_{\lambda,u}(\theta),$$

is decreasing in u. When $u = 0$, no budget penalty and $\hat{\rho}_{\lambda,0}$ concentrates on model/parameter space regions with high empirical welfare as $\lambda \to \infty$.

Two perspectives:

A. Choose $\hat{u}$ to ensure $B_n(\hat{\rho}_{\lambda,\hat{u}}) \leq B$.

B. Choose $u \in \mathcal{W}_u$. Range of u values, estimate of budget for each.

For scenario B, for simplicity assume $|\mathcal{W}_u| = n$.

Motivating the Gibbs Rules

Motivation 1: Stochastic methods are of general interest

- Suppose an estimated stochastic treatment rule is such that we know the treatment probabilities and treatments are randomized on observable characteristics.
- Provided strict overlap holds, after treating people, the newly treated population is effectively another sample of individuals who have received a randomized treatment. Can facilitate analysis on policy effectiveness, potentially provide path to re-estimating a treatment policy with a larger sample.
- Finding methods that assign treatment in such a manner, and do so with desirable properties, is of general interest. Gibbs treatment rules will often fit the bill.

Motivation 2: Alteration to a nice optimization problem

Restricting attention to $\mathcal{F}_\Theta$, empirical counterpart of policy problem is

$$\max_{\theta \in \mathbb{R}^q} \frac{1}{n} \sum_{i=1}^{n} \delta_{y,i}(X_i) 1\{\phi(X_i)'\theta > 0\} \qquad (\max_\theta W_n(\theta))$$

$$\text{s.t. } \frac{1}{n} \sum_{i=1}^{n} \delta_{c,i}(X_i) 1\{\phi(X_i)'\theta > 0\} \leq B \qquad (\text{s.t. } K_n(\theta) \leq B)$$

Note: Objective function is not generally concave in θ and the set of θ in $\mathbb{R}^q$ that satisfy the constraint is not generally convex. Problems:

- Difficult to analyze a solution $\hat{\theta}$ theoretically.
- Computationally difficult to find a solution $\hat{\theta}$.
- [Kitagawa et al., 2021]: in settings with constraints, surrogate loss methods can lose properties like Fisher consistency (when the solution to a minimization problem with a surrogate loss function yields the same decisions as the original optimization problem).

Alteration to a nice optimization problem

$$D_{\mathrm{KL}}(\rho, \pi) = \begin{cases} \int \log\left[\frac{d\rho}{d\pi}(\theta)\right] d\rho(\theta), & \text{if } \rho \ll \pi \\ \infty, & \text{else.} \end{cases}$$

Policy maker's problem considering Gibbs rules is $\max_\rho W(f_{G,\rho})$ s.t. $K(f_{G,\rho}) \leq B$, i.e.

$$\max_{\rho \in \mathcal{P}(\Theta)} \int W(\theta) d\rho(\theta), \text{ s.t. } \int K(\theta) d\rho(\theta) \leq B.$$

Alteration to a nice optimization problem

We will instead focus on a subset $\mathcal{P}_\pi(\Theta) \subset \mathcal{P}(\Theta)$ and solve the following penalized empirical problem:

$$\max_{\rho \in \mathcal{P}_\pi(\Theta)} \left[\int W_n(\theta) d\rho(\theta) - \frac{1}{\lambda} D_{\mathrm{KL}}(\rho, \pi) \right] \quad \text{s.t.} \int K_n(\theta) d\rho(\theta) \leq B.$$

where $\mathcal{P}_\pi(\Theta)$ denotes all ρ such that $\rho \ll \pi$.

- Objective is concave in ρ, convex set of ρ satisfying constraint.
- Can use Bayesian estimation methods to utilize $\hat{\rho}_{\lambda,u}$ (solution).
- Utilize dual problem in theoretical analysis, enables theoretical analysis.
- Sharp oracle inequalities that relate solution to original population-level policy maker problem over $\mathcal{F}_\Theta$.

Constraint Qualification Assumption QCA

(a) Let $B \in \mathbb{R} \cup \{\infty\}$.

$$\pi\left(\theta : K(\theta) < B\right) > 0 \;\text{ and }\; \pi\left(\theta : K_n(\theta) < B\right) > 0 \;\text{ almost surely.}$$

(b)

$$\mathbb{V}_{\theta\sim\pi}\left[K(\theta)\right] > 0 \;\text{ and }\; \mathbb{V}_{\theta\sim\pi}\left[K_n(\theta)\right] > 0 \;\text{ almost surely}$$

where, $\mathbb{V}_{\theta\sim\pi}$ denotes the variance of $K(\theta)$ when $\theta \sim \pi$ and, for a fixed sample $S \in \mathcal{S}$, $\mathbb{V}_{\theta\sim\pi}[K_n(\theta)]$ denotes the variance of $K_n(\theta)$ when $\theta \sim \pi$.

Definitions and lemma

$$\hat{\mathcal{E}}_B = \{\rho \in \mathcal{P}_\pi(\Theta) : B_n(\rho) \leq B\},$$

$\hat{u}(B, \lambda)$: $\hat{u}$ s.t. $B_n(\hat{\rho}_{\lambda,\hat{u}}) = B$ if $B_n(\hat{\rho}_{\lambda,0}) > B$, else $\hat{u}(B, \lambda) = 0$.

Lemma:

Under the CQA (a), $\hat{u}(B, \lambda) \geq 0$ exists and is unique. Also, w/ prob. one,

$$\hat{\rho}_{\lambda,\hat{u}(B,\lambda)} = \arg\max_{\hat{\mathcal{E}}_B} \left[\int W_n(\theta) d\rho(\theta) - \frac{1}{\lambda} D_{\mathrm{KL}}(\rho, \pi)\right],$$

and

$$\max_{\rho \in \mathcal{P}_\pi(\Theta)} \inf_{u \geq 0} \left[\int W_n(\theta) d\rho(\theta) - \frac{1}{\lambda} D_{\mathrm{KL}}(\rho, \pi) - u\left(\int K_n(\theta) d\rho(\theta) - B\right)\right]$$
$$= \inf_{u \geq 0} \max_{\rho \in \mathcal{P}_\pi(\Theta)} \left[\int W_n(\theta) d\rho(\theta) - \frac{1}{\lambda} D_{\mathrm{KL}}(\rho, \pi) - u\left(\int K_n(\theta) d\rho(\theta) - B\right)\right]$$

arg max in r.h.s. is $\hat{\rho}_{\lambda,u}$. For fixed $u \geq 0$, counterparts of the above hold with $\hat{u}(B, \lambda)$ replaced by u, B replaced by $B_n(\hat{\rho}_{\lambda,u})$ and CQA (b) in place of (a).

Theoretical Properties

Simplest PAC-Bayesian generalization bound for our setting

Adapting the formula/approach laid out in, for example, in [Catoni, 2007], [Alquier et al., 2016], and [Germain et al., 2015],

Lemma: PAC-Bayesian generalization bound

With probability at least $1-\epsilon$, it holds for all $\rho \ll \pi$ simultaneously that

$$W(f_{G,\rho}) \geq W_n(f_{G,\rho}) - \frac{1}{\lambda} D_{\mathrm{KL}}(\rho, \pi) - \frac{\lambda M_y^2}{8n\kappa^2} - \frac{1}{\lambda} \log \frac{1}{\epsilon}.$$

proof

- Similar can be derived for $K(f_{G,\rho})$ and $K_n(f_{G,\rho})$ with inequality flipped and $+$'s instead of $-$'s. $K_n(f_{G,\hat{\rho}_{\lambda,\hat{u}(B,\lambda)}}) \leq B$.
- $\hat{\rho}_{\lambda,u}$ maximizes lower bound on the welfare over $\hat{\mathcal{E}}_{B_n(\hat{\rho}_{\lambda,u})}$.
- Expert knowledge informing π: smaller $D_{\mathrm{KL}}(\rho, \pi)$ for Gibbs rules associated with ρ that yield better welfare, larger lower bound.

Generalization bounds for cost, $u \in \mathcal{W}_u$

Theorem: generalization bound for costs, $u \in \mathcal{W}_u$

Fix $\epsilon \in (0, 1]$. With probability at least $1 - \epsilon$, for all $u \in \mathcal{W}_u$ simultaneously it holds that

$$|B(\hat{\rho}_{\lambda,u}) - B_n(\hat{\rho}_{\lambda,u})| \leq U_c(\epsilon; \lambda, u, n).$$

If $\lambda = \mathcal{O}(\sqrt{n})$ and $\max_u \mathcal{W}_u = \mathcal{O}(1)$, then

$$\max_{u \in \mathcal{W}_u} U_c(\epsilon; \lambda, u, n) = \mathcal{O}\left(\frac{\log n}{\sqrt{n}}\right).$$

Explicit form for $U_c(\epsilon; \lambda, u, n)$ in the paper.

– Can choose $u \in \mathcal{W}_u$ in data dependent manner.

– New result, similar to a bound in [Lever et al., 2010] and proof utilizes the approach there.

Oracle inequalities: optimal rules & welfare regret

Define CATE, CATC: $\delta_y(x) = E[Y_1 - Y_0|X = x]$, $\delta_c(x) = E[C_1 - C_0|X = x]$.

Lemma: optimal treatment rules

Under $B > E[\delta_c(X)1\{\delta_c(X) < 0\}]$ and regularity conditions, solution to $\max_f W(f)$ s.t. $K(f) \leq B$ is:

$$f_B^*(x) = 1\{\delta_y(x) \geq \eta_B \delta_c(x)\},$$

for some constant η_B that depends on B. $\eta_\infty = 0$, denote $f_\infty^*(x) = f^*(x)$.

Definition: (non-negative) welfare regret

$$R(f) = W(f^*) - W(f); \quad R(\theta) = R(f_\theta) \text{ for } f_\theta \in \mathcal{F}_\Theta$$

Oracle inequalities: parameter sets of interest

$$\Theta_B = \{\theta \in \mathbb{R}^q : K(\theta) \leq B\}\,; \quad \Theta_{B(\hat{\rho}_{\lambda,u})} = \{\theta \in \mathbb{R}^q : K(\theta) \leq B(\hat{\rho}_{\lambda,u})\}\,.$$

Assumption With probability one,

$$\overline{\theta} \in \underset{\Theta_B}{\arg\min}\,[R(\theta)] \text{ and } \overline{\theta}_u \in \underset{\Theta_{B(\hat{\rho}_{\lambda,u})}}{\arg\min}\,[R(\theta)]\,,$$

exist and are nonzero.

–Existence of solution is standard implicit assumption in EWM (e.g. [Kitagawa et al., 2021] and [Sun, 2021]).

–Nonzero: covariates are relevant.

Oracle inequalities

Assumption There exists a constant $\nu > 0$ such that

$$P\left[(\phi(X)^\intercal\theta)\left(\phi(X)^\intercal\theta'\right) < 0\right] \leq \nu\|\theta - \theta'\|$$

for any θ and $\theta' \in \mathbb{R}^q$ such that $\|\theta\| = \|\theta'\| = 1$.

- Probability that a small change in θ will change the treatment rule $f_\theta(X) = 1\{\phi(X)^\intercal\theta > 0\}$ is low.
- Mild. [Alquier et al., 2016]: satisfied whenever $\phi(X)/||\phi(X)||$ has bounded density on the unit sphere. Direct analogue of assumptions in several PAC-Bayesian classification and bipartite ranking papers, e.g. [Ridgway et al., 2014] and [Alquier et al., 2016].

Theorem. Take π as $N(0, I_q/\sqrt{q})$, fix $\epsilon \in (0, 1]$.

For pre-determined B

Let $\lambda = \kappa\sqrt{nq}/M_y$, $\hat{u} = \hat{u}(B, \lambda)$, and $u^* = u^*(B, \lambda/2)$ (c.f. next slide). With probability at least $1 - \epsilon$,

$$R\left(f_{G,\hat{\rho}_{\lambda,\hat{u}}}\right) \leq R\left(\overline{\theta}\right) + \hat{u}\sqrt{\frac{M_c^2 \log \frac{3}{\epsilon}}{2n\kappa^2}} + \frac{u^* \nu M_c}{\sqrt{n}} + \overline{U}_1(\epsilon; n, q)$$

where $\overline{U}_1(\epsilon; n, q) = \mathcal{O}(\log(n) n^{-1/2})$.

For $u \in \mathcal{W}_u$

Let $\lambda = \kappa\sqrt{nq}/(M_y + uM_c)$. With probability at least $1 - \epsilon$, for all $u \in \mathcal{W}_u$ simultaneously it holds that

$$R\left(f_{G,\hat{\rho}_{\lambda,u}}\right) \leq R\left(\overline{\theta}_u\right) + \frac{M_y + uM_c}{\kappa}\left[\overline{U}_2(\epsilon; u, n, q)\right].$$

When $\max_u \mathcal{W}_u = \mathcal{O}(1)$, $\max_{u \in \mathcal{W}_u} \overline{U}_2(\epsilon; u, n, q) = \mathcal{O}(\log(n) n^{-1/2})$.

Oracle inequalities: notes

- $u^*(B, \lambda)$ is defined the same way as $\hat{u}(B, \lambda)$ but with $W(\theta)$ and $K(\theta)$ in place of $W_n(\theta)$ and $K_n(\theta)$. $\rho^*_{\lambda,u}$ has RN-derivative prop. to

$$\exp\left[-\lambda\left(uK(\theta) - W(\theta)\right)\right].$$

  $u^*(B, \lambda)$ is the value of u^* such that $B(\rho^*_{\lambda,u^*}) = B$ if $B(\rho^*_{\lambda,0}) > B$ or else $u^*(B, \lambda) = 0$.

- λ were chosen for $\log(n)n^{-1/2}$ rate of convergence. In practice there may be better choices: we choose λ via cross-validation. Provided $|\mathcal{W}_u \times \mathcal{W}_\lambda| = cn$ for some constant c and $\lambda = \mathcal{O}(n^{1/2})$, will not effect rates in theorem.

idea behind the proof

Simulation Study

Simulation setup

- $X = (X_1, X_2, X_3)$, $X_j \sim \mathrm{Unif}(-1, 1)$ for $j = 1, 2, 3$ are i.i.d.
- With logistic function $\Lambda(v) = (1 + \exp(-v))^{-1}$,

  $$Y_d = \max\{X_1 + X_2, 0\} + \max\{X_3, 0\} + 4d\Lambda\left(2\left(X_1 + X_1X_2 + X_2\right)\right) + \epsilon, \quad d \in \{0, 1\},$$

  where ϵ is a standard normal, truncated to $[-2, 2]$, independent of other variables.
- Potential costs:

  $$C_0 = 0, \quad C_1 \sim \mathrm{Binom}\left(6, \frac{4\Lambda\left(a(3X_2 + 1.5X_3)\right)}{6}\right), \quad a \in \{1, 2, 4\}.$$
- $e(x) = 1/2$ for all $x \in \mathcal{X}$, $D \sim \mathrm{Bern}(1/2)$, independent of other variables.

Models considered

- $f_{G,\hat{\rho}_{\lambda,u}}$, denoted PB-G, and $f_{\mathrm{mv},\hat{\rho}_{\lambda,u}}$, denoted PB-MV.
- $\mathcal{W}_u = \{0, 0.05, 0.1, \ldots, 2\}$. Cross validate λ for each to obtain PB-G and PB-MV model for each u.
- For $k \in \mathbb{N}$ and $q_k = \binom{d_x+k}{k}$, polynomial transformation on $\mathbb{R}^{d_x}$ is

$$\mathcal{F}_{\Theta}^{\mathrm{poly}}(k) = \left\{ m(x) : m(x) = \sum_{j=1}^{q_k} \theta_j \phi_j(x), \theta \in \mathbb{R}^{q_k} \right\}. \tag{1}$$

  Summation is over all monomials $\phi_j(x) = \prod_{\ell=1}^{d} x_\ell^{p_{j\ell}}$ with $\sum_{\ell=1}^{d} p_{j\ell} \leq q$, $p_{j\ell} \in \mathbb{N} \cup \{0\}$.
- Take $\phi_j(X)$, $j = 1, \ldots, 10$, in $f_\theta(x) = 1\{\phi(X)^\intercal \theta > 0\}$ to be the monomials in the polynomial transformation on $\mathbb{R}^3$ of order at most 2 standardized by their sample means and standard deviations. Take π to be the standard normal distribution over $\mathbb{R}^{10}$.

Models considered

To sample from $\hat{\rho}_{\lambda,u}$ to compute $\int f_\theta(X_i) d\hat{\rho}_{\lambda,u}(\theta)$ and implement Gibbs or majority vote rules, we can utilize Bayesian estimation techniques. In paper, utilize the Sequential Monte Carlo (SMC) approach of [Del Moral et al., 2006].

Batch treatment method: when costs are recognized as treatments are assigned, a treatment method will be called a batched treatment method if it assigns some sort of score value to each individual in the target group (based on their covariate values) and then assigns treatment in descending order of the score value until the desired budget level is exhausted. The Gibbs and majority vote methods are not batch treatment methods.

Models considered

- Alternative: When $C_0 \leq C_1$ a.s., for feasible B, optimal rule takes the form

$$f_B^*(x) = 1\left\{\frac{\delta_y(x)}{\delta_c(x)} > \eta_B\right\}.$$

  [Sun et al., 2021] show the ratio can be estimated using the generalized random forest method of [Athey et al., 2019]. This estimate yields an estimated conditional welfare gain to cost ratio score for each member of the target group, who are then ordered in descending order of score. Then for a desired cost level, treatments are assigned via a batch implementation. Denoted R-NP.

- Baseline: estimate $\delta_y(x)$ using the generalized random forest method, batch implementation similarly to the [Sun et al., 2021] methodology.

Models considered: batch implementation via $f_{\mathrm{MV},\hat{\rho}_{\lambda,u}}$ or $f_{G,\hat{\rho}_{\lambda,u}}$

For a set of u values, we obtain $f_{\mathrm{MV},\hat{\rho}_{\tilde{\lambda}_u,u}}$ and $\hat{\mathrm{cost}}(f_{\mathrm{MV},\hat{\rho}_{\tilde{\lambda}_u,u}})$ for each (for example during CV we get a cost estimate). A batch implementation utilizing the resulting $\hat{\rho}_{\tilde{\lambda}_u,u}$ distributions is as follows:

1. For a desired budget level B, divide $[0, B]$ into n_{bin} equally spaced bins.
2. Starting from the bin nearest to 0, at end point $\frac{1 \cdot B}{n_{bin}}$, identify $u \in \mathcal{W}_u$ such that $\hat{\mathrm{cost}}(f_{\mathrm{MV},\hat{\rho}_{\tilde{\lambda}_u,u}})$ is closest to $\frac{1 \cdot B}{n_{bin}}$. Then rank target group observations by the scores $\int f_\theta(X_i) d\hat{\rho}_{\tilde{\lambda}_u,u}(\theta)$ and implement a batch implementation until the cost $\frac{1 \cdot B}{n_{bin}}$ is reached.
3. Do the same process at subsequent bin ends $\frac{j \cdot B}{n_{bin}}$, $j = 2, \ldots, n_{bin}$ only considering target group individuals who have not yet been treated in the rankings, until the budget B is reached or all individuals are treated.

Denote PB-B.

Simulation details

For each a, we do the following simulation setup:

- Training sets with sample size $n = 1,000$
- Test set of size $n_{\text{test}} = 10,000$. Using knowledge of DGP, calculate $E[Y_{1,i} - Y_{0,i}|X_i]$ and $E[C_{1,i}|X_i]$ for each observation in the test set. Then for a rule f, via the test sample we can approximate

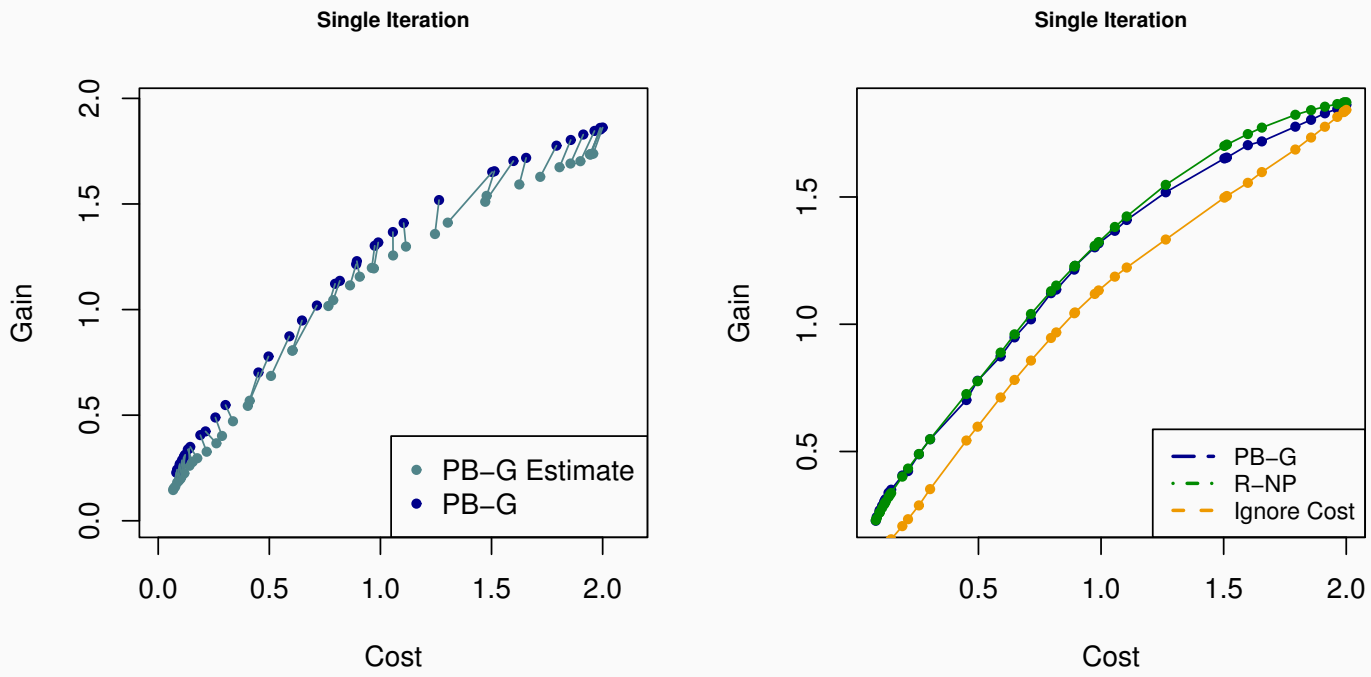
  $$\text{Gain of } f = E[(Y_1 - Y_0)f(X)]$$

  and

  $$\text{Cost of } f = E[C_1 f(X)]$$

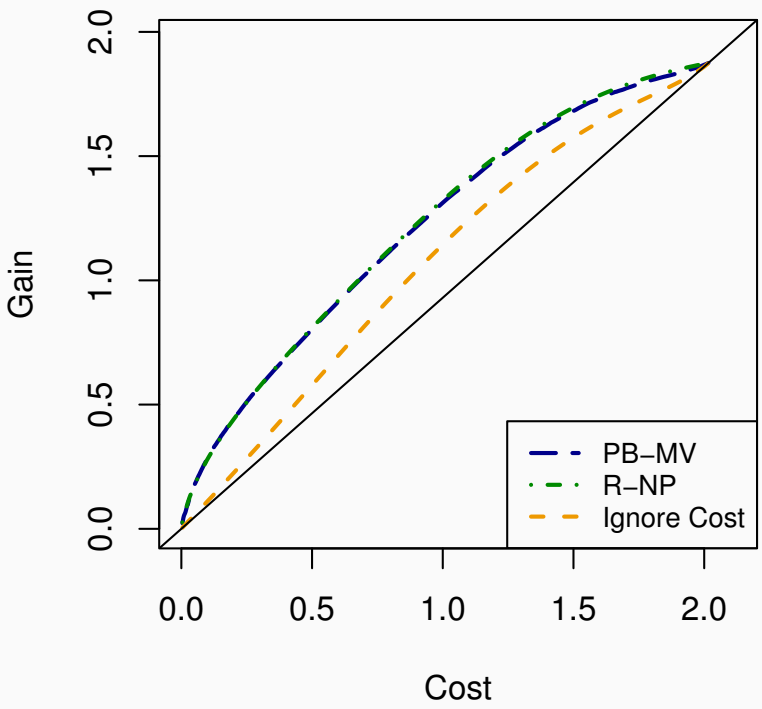
- Consider 100 training sample replicates.
- Compute cost curves: illustrated on subsequent slides.

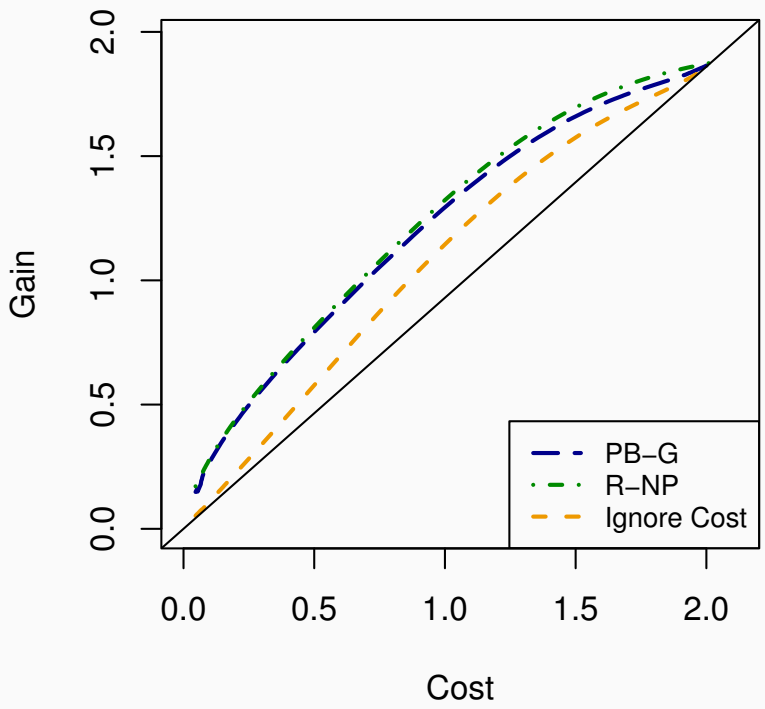
Cost Curves: a single iteration for PB-G when $a = 1$ in DGP



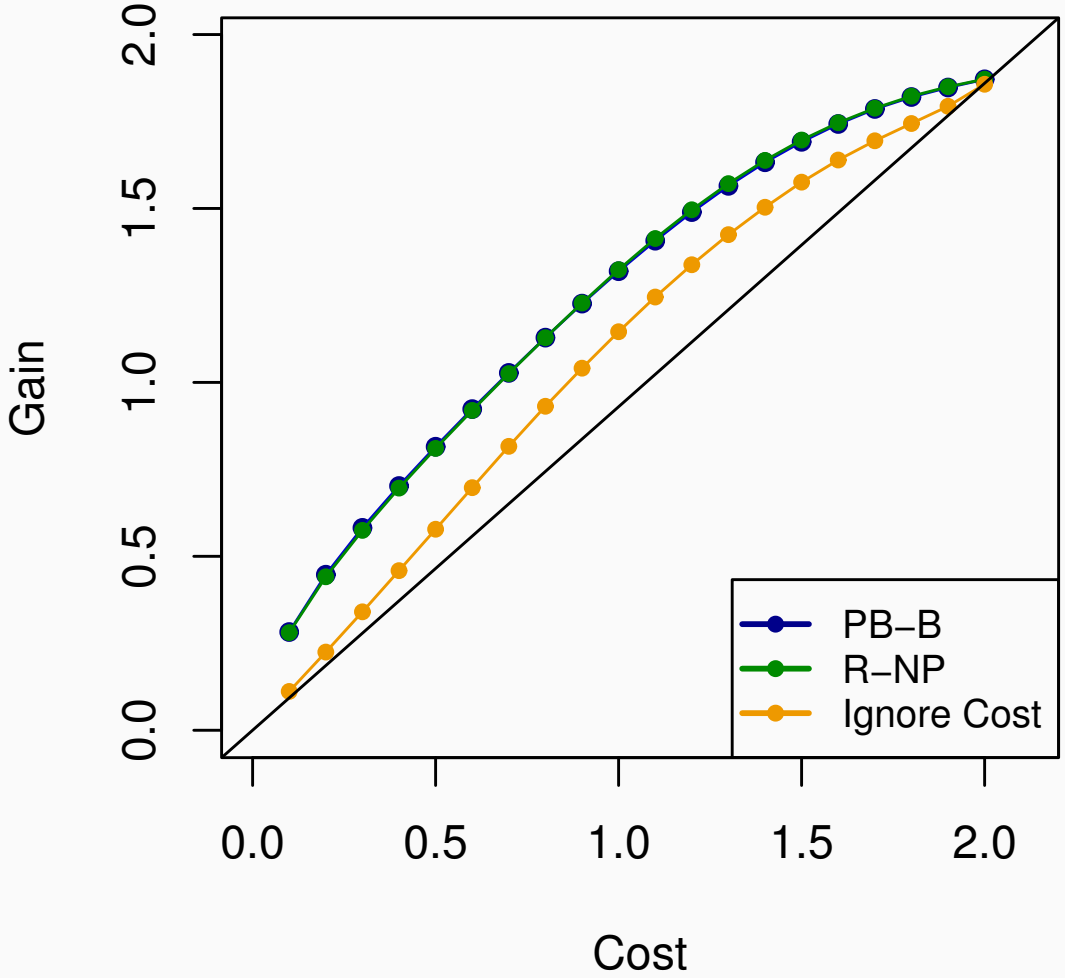
Left: the estimated and actual cost-gain pairs (one point for each u) for a single training sample iteration for PB-G. Right: the actual cost-gain pairs for PB-G for various u values are plotted again, now compared with the actual cost-gain pairs associated with the R-NP and IC rules that produce the same target population cost. The points on the right are interpolated to produce cost-gain curve estimates for a single iteration. These curves are averaged (vertically) over all simulation iterations (on next slide).

Simulation results: cost curves when $a = 1$, PB-G & PB-MV vs batch alternatives

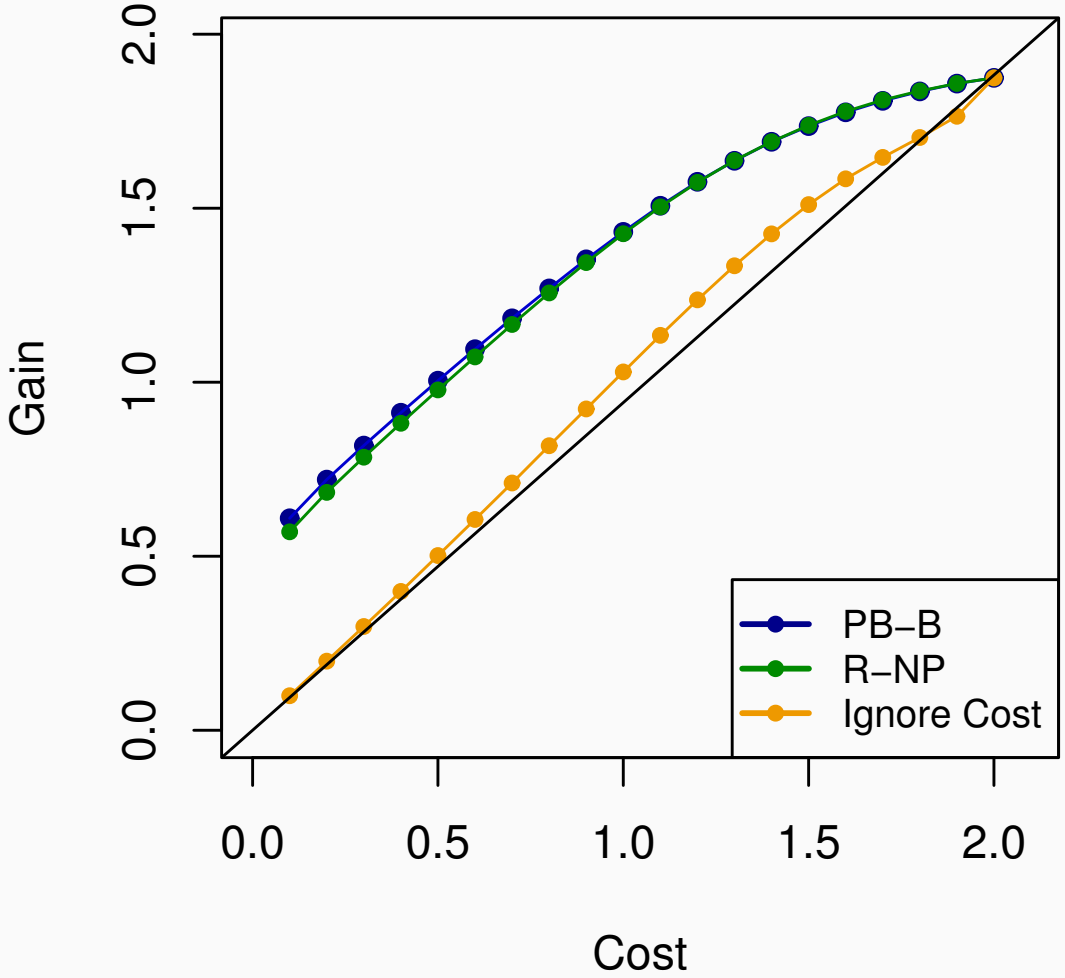




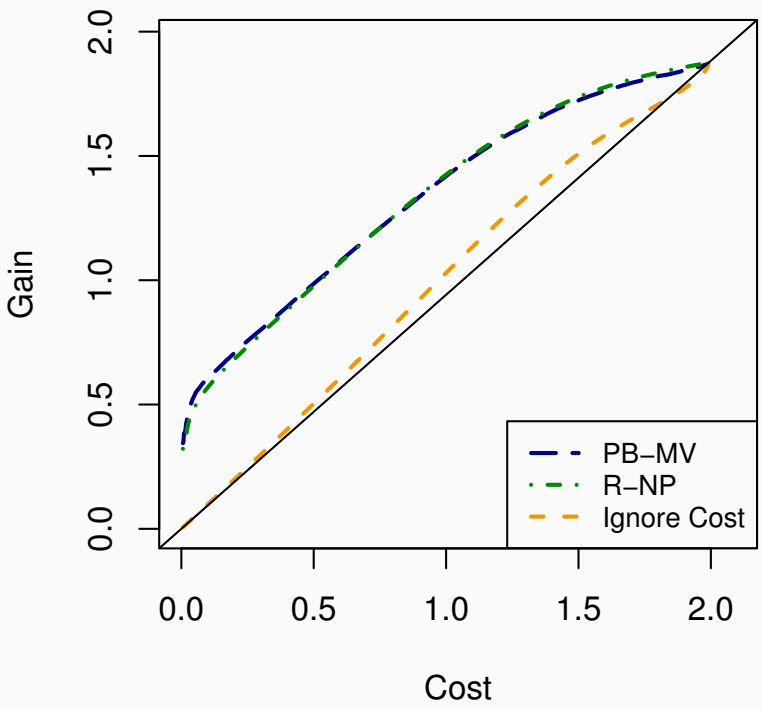
Simulation results: cost curves when $a = 1$, batch methods

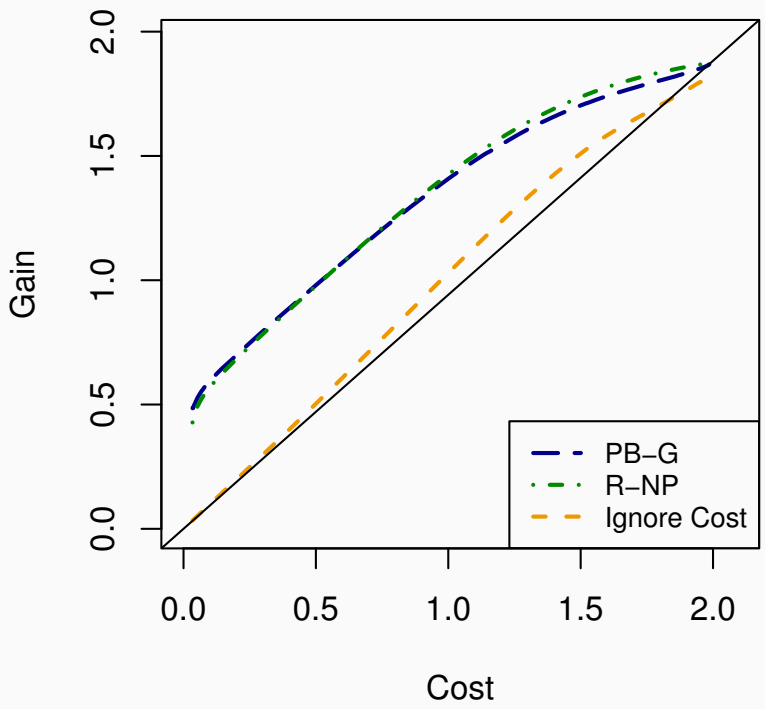


Simulation results: cost curves when $a = 4$, batch methods



Simulation results: cost curves when $a = 4$, PB-G & PB-MV vs batch alternatives





Simulation results

Cost	PB-G	PB-MV	PB-B	R-NP	Ignore Cost
			$a = 1$		
0.1	0.27	0.28	0.28	0.28	0.11
0.3	0.56	0.57	0.58	0.57	0.34
0.6	0.90	0.91	0.92	0.92	0.70
0.9	1.20	1.22	1.23	1.23	1.04
1.2	1.46	1.48	1.49	1.49	1.34
1.5	1.66	1.68	1.69	1.70	1.57
1.8	1.79	1.81	1.82	1.82	1.74
			$a = 2$		
0.1	0.46	0.47	0.47	0.45	0.10
0.3	0.71	0.72	0.73	0.70	0.31
0.6	1.01	1.01	1.03	1.01	0.63
0.9	1.28	1.29	1.30	1.29	0.96
1.2	1.51	1.53	1.54	1.54	1.26
1.5	1.68	1.70	1.71	1.72	1.52
1.8	1.79	1.81	1.82	1.83	1.71
			$a = 4$		
0.1	0.60	0.61	0.61	0.57	0.10
0.3	0.80	0.80	0.82	0.78	0.30
0.6	1.07	1.08	1.09	1.07	0.61
0.9	1.33	1.34	1.35	1.34	0.92
1.2	1.55	1.56	1.58	1.57	1.24
1.5	1.70	1.72	1.74	1.74	1.51
1.8	1.80	1.82	1.84	1.83	1.70

Simulation results: heuristic reflection on this example

- $\delta_y(x) = 4\Lambda(2(x_1 + x_1x_2 + x_2))$, $\delta_c(x) = 4\Lambda(a(3x_2 + 1.5x_3))$, so that

$$f_B^* = 1\{\delta_y(x)/\delta_c(x) > \eta_B\} = 1\{\delta_y(x) > \eta_B\delta_c(x)\}.$$

  For low cost models (B small), η_B is larger.
- Near the origin, $\Lambda(x)$ is approximately linear and the conditional welfare to cost ratio is simpler. As a increases and for smaller B, we are more likely to need to make decisions based on regions of the covariate space where this approximation is less accurate and the optimal rule / conditional welfare to cost ratio is a more complicated object.
- In areas where we may be more uncertain about the model estimates, the PAC-Bayesian-based aggregation approaches that average over models with similar empirical welfare that might disagree perform better than when we pick a single rule in this example.

Empirical Illustration

Empirical illustration: Gibbs rules for the JTPA Study data

- The Job Training Partnership Act (JTPA) Study was a randomized controlled trail aimed at assessing the benefits/costs of training and employment assistance programs of the Job Training Partnership Act. Detailed description in [Bloom et al., 1997].
- Treatment group: eligibility for variety of JTPA services. Control group: restricted access (e.g. ineligibility period for most services). Treatment is *eligibility* not use/compliance.
- Treatment costs vary due to whether services were utilized and which types. Among control group members who received services, costs similar to treatment group individuals that received services. Probability of not receiving any services the key driver of cost differences [Orr et al., 1994].

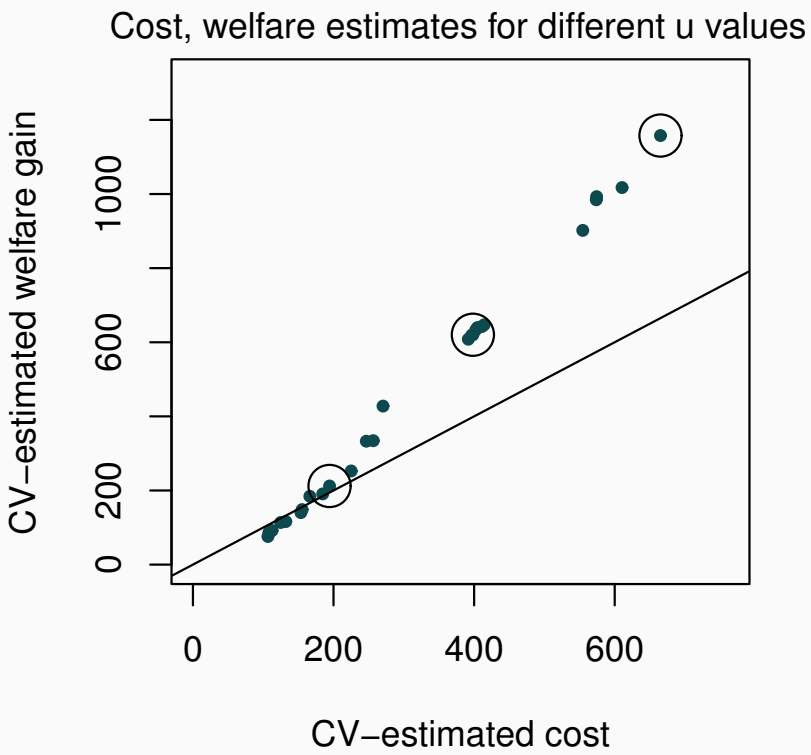
Empirical illustration: sample, cost calculations

- 7,675 observations among individuals used in original program evaluation (e.g. [Bloom et al., 1997])) consisting of adults (22 years and older) for whom data on years of education, pre-program earnings, and whether or not any treatment services were utilized was available. The probability of being assigned treatment was 2/3 in this sample.
- Y is total individual earnings in the 30 months following treatment assignment.
- X are an individual's years of education and earnings in the year prior to treatment assignment.
- Cost measurement: 0 if the individual utilized no JTPA services.
  [Orr et al., 1994] report average costs for subgroups based on gender, treatment vs control group assignment, and service recommendation (Classroom training, OJT/JSA, Other Services; recommendation correlated with services used). For individuals who utilized > 0 hours of JTPA services, we assigned their cost as the mean of their group according to these characteristics.

Empirical illustration: modeling decisions

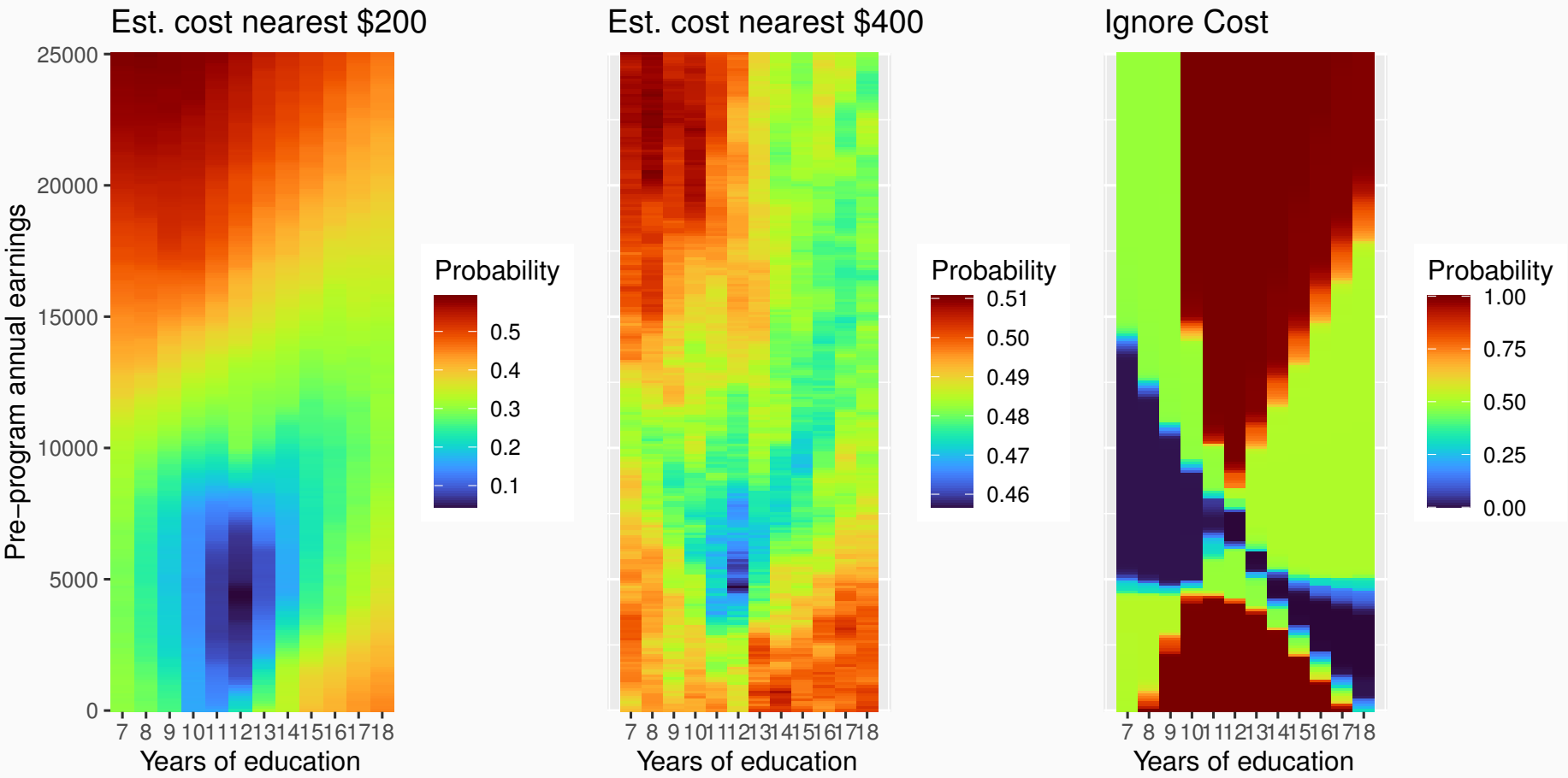
- $\phi_1(x), \ldots, \phi_{10}(x)$: monomials from the polynomial transformations on $(edu, earn)$ space of order at most 3. Normalized as in the simulation section (education ranges from 7 to 18 years, pre-program earnings range from \$0 to \$63000).
- Estimation via SMC. Consider $u \in \{0.05, 0.1, \ldots, 3\}$. λ: 2-fold cross validated. Consider $\lambda \in \{2, 2^2, \ldots, 2^{10}\}/(1+u)$ for each u. π: std. multivariate normal over $\mathbb{R}^{10}$.

CV-based est. costs for Gibbs treatment rules with different u

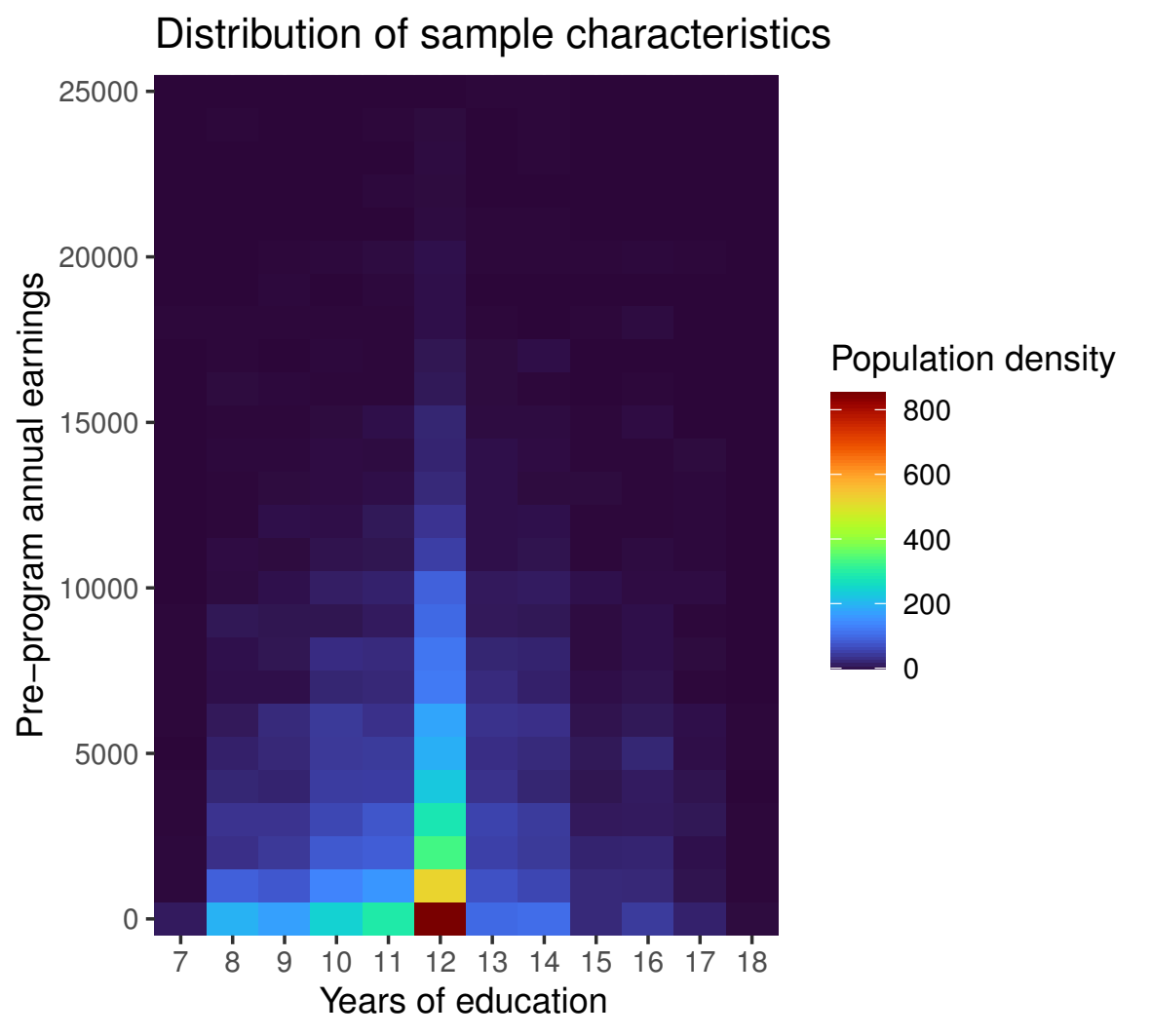


For each u, abusing notation, each hold-out sample (fold) during CV yields estimates $W_n(f_{G,\hat{\rho}_{\lambda,u}})$ and $B_n(\hat{\rho}_{\lambda,u})$ calculated on that fold. Averaging across folds gives the CV estimates for each λ. A dot above corresponds to the hold-out fold averages of $W_n(f_{G,\hat{\rho}_{\tilde{\lambda}_u,u}})$ and $B_n(\hat{\rho}_{\tilde{\lambda}_u,u})$ for the $\tilde{\lambda}_u$ selected by CV.

Treatment probabilities for models with different costs



Population density



Notes:

- Treatment probabilities are not uniform. Lower is not achieved by simply lowering probability uniformly versus higher cost model.
- Treatment probabilities vary fairly smoothly across the covariate space (fairness).
- The EWM rules in [Kitagawa and Tetenov, 2018] mostly avoid treating above a certain education level. Here, the rules do assign some probability to high education levels, particularly lower cost models.
- No budget model would on avg treat 81% of individuals, medium budget treats 49% and treatment probabilities among observed covariates range from 46% to 51%, low-budget treats 20% of individuals and treatment probabilities range from 4% to 66% among covariates observed in sample.
- As we target rules with lower (average) cost, we assign lower treatment probabilities to individuals with middling to low (but not zero) pre-program earnings and middle to low education levels.

End

Proof of simplest PAC-Bayesian generalization bound

- Change-of-measure ineq. [Csiszar, 1975, Donsker & Varadhan, 1975]:

$$\forall \rho \text{ on } (\mathbb{R}^q, \mathcal{B}(\mathbb{R}^q)), \int_\Theta \lambda \left[W_n(\theta) - W(\theta)\right] d\rho(\theta)$$
$$\leq \log \left(\int_\Theta \exp\left(\lambda \left[W_n(\theta) - W(\theta)\right]\right) d\pi(\theta) \right) + D_{\mathrm{KL}}(\rho, \pi).$$

- Let $\theta \in \mathbb{R}^q$ and $\lambda > 0$. Manipulations w/ Hoeffding's Lemma produce,

$$E[\exp\left(\lambda \left[W_n(\theta) - W(\theta)\right]\right)] \leq \exp\left(\frac{\lambda^2 M_y^2}{8n\kappa^2}\right).$$

back

Proof of simplest PAC-Bayesian generalization bound

Markov's inequality for $\epsilon \in (0, 1]$ and non-negative r.v. Q:

$$P\left[\log Q \leq \log E[Q] + \log(1/\epsilon)\right] \geq 1 - \epsilon$$

Set $Q = \int \exp[\lambda(W_n(\theta) - W(\theta))]d\pi(\theta)$. With probability at least $1 - \epsilon$,

$$\begin{aligned}
&\log\left(\int \exp(\lambda[W_n(\theta) - W(\theta)])d\pi(\theta)\right) \\
&\leq \log E\left[\int \exp[\lambda(W_n(\theta) - W(\theta))]d\pi(\theta)\right] + \log(1/\epsilon) \\
&= \log \int E \exp[\lambda(W_n(\theta) - W(\theta))]d\pi(\theta) + \log(1/\epsilon) && \text{(Fubini)} \\
&\leq \frac{\lambda^2 M_y^2}{8n\kappa^2} + \log(1/\epsilon). && \text{(MGF u.b.)}
\end{aligned}$$

back

Idea behind proof of oracle inequality for $u \in \mathcal{W}_u$

Derived from corresponding theorem for Gibbs rules.

Letting

$$\mathcal{E}_B = \{\rho \in \mathcal{P}_\pi(\Theta) : B(\rho) \leq B\},$$

the theorem is derived from the starting point of showing: w/ prob. at least $1 - \epsilon$,

$$R(f_{G,\hat{\rho}_{\lambda,u}}) \leq \min_{\rho \in \mathcal{E}_B(\hat{\rho}_{\lambda,u})} \left\{ R(f_{G,\rho}) + \frac{1}{\lambda} D_{\mathrm{KL}}(\rho, \pi) \right\} + \overline{U}(\epsilon, \lambda, n, u)$$

where $\overline{U}(\epsilon, \lambda, n, u)$ is a remainder term that gets small when $\lambda = \mathcal{O}(n^{1/2})$.

back

Idea behind proof of oracle inequality for $u \in \mathcal{W}_u$

Here: give idea behind the proof of the high probability inequality last page.

Letting

$$R_n(f) = \frac{1}{n} \sum_{i=1}^{n} \delta_{y,i}(f^*(X_i) - f(X_i)),$$

and, as with earlier notation, $R_n(\theta) = R_n(f_\theta)$. Because f^* does not vary with θ,

$$\frac{d\hat{\rho}_{\lambda,u}}{d\pi}(\theta) = \frac{\exp\left[-\lambda\left(R_n(\theta) + uK_n(\theta)\right)\right]}{\int \exp\left[-\lambda\left(R_n\left(\tilde{\theta}\right) + uK_n\left(\tilde{\theta}\right)\right)\right] d\pi\left(\tilde{\theta}\right)}.$$

Similarly, $\rho^*_{\lambda,u}$ is equivalent to the above but with $R_n(\theta)$ and $K_n(\theta)$ replaced by $R(\theta)$ and $K(\theta)$, respectively.

back

Idea behind proof of oracle inequality for $u \in \mathcal{W}_u$

Alternative version of earlier Lemma

Under CQA, w/ prob. one,

$$\min_{\mathcal{E}_{B(\hat{\rho}_{\lambda,u})}} \left[\int R(\theta) d\rho(\theta) + \frac{1}{\lambda} D_{\mathrm{KL}}(\rho, \pi) \right]$$

$$= \min_{\rho \in \mathcal{P}_\pi(\Theta)} \sup_{u \geq 0} \left[\int R(\theta) d\rho(\theta) + \frac{1}{\lambda} D_{\mathrm{KL}}(\rho, \pi) + u \left(\int K(\theta) d\rho(\theta) - B(\hat{\rho}_{\lambda,u}) \right) \right]$$

$$= \sup_{u \geq 0} \min_{\rho \in \mathcal{P}_\pi(\Theta)} \left[\int R(\theta) d\rho(\theta) + \frac{1}{\lambda} D_{\mathrm{KL}}(\rho, \pi) + u \left(\int K(\theta) d\rho(\theta) - B(\hat{\rho}_{\lambda,u}) \right) \right]$$

back

Idea behind proof of oracle inequality for $u \in \mathcal{W}_u$

In paper, show the following hold with prob. $1 - \epsilon$ each:

$$R(f_{G,\hat{\rho}_{\lambda,u}}) \leq R_n(f_{G,\hat{\rho}_{\lambda,u}}) + \frac{1}{\lambda} D_{\mathrm{KL}}(\hat{\rho}_{\lambda,u}, \pi) + U_*(\epsilon; \lambda, n), \tag{*}$$

and

$$\begin{aligned} &\int R_n(\theta) d\hat{\rho}_{\lambda,u}(\theta) + u \int K_n(\theta) d\hat{\rho}_{\lambda,u}(\theta) + \frac{1}{\lambda} D_{\mathrm{KL}}\left(\hat{\rho}_{\lambda,u}, \pi\right) \\ &\leq \int R(\theta) d\rho^*_{\lambda,u}(\theta) + u \int K(\theta) d\rho^*_{\lambda,u}(\theta) + \frac{1}{\lambda} D_{\mathrm{KL}}(\rho^*_{\lambda,u}, \pi) + U_{**}(\epsilon; u, n), \end{aligned} \tag{**}$$

$U_*(\epsilon; \lambda, n)$ and $U_{**}(\epsilon; u, n)$ are given in paper. We can combine separate probabilistic inequalities with a union bound argument (the combination will have remainder terms that are slightly larger than $U_*(\epsilon; \lambda, n)$ and $U_{**}(\epsilon; u, n)$ via $1/\epsilon$ terms replaced with $2/\epsilon$).

Recall the definition $B_n(\hat{\rho}_{\lambda,u}) = \int K_n(\theta) d\hat{\rho}_{\lambda,u}(\theta)$.

Idea behind proof of oracle inequality for $u \in \mathcal{W}_u$

Below, $U(\epsilon; \lambda, u, n)$ is a catchall term for remainders, these are dealt with in the paper but not essential to idea behind proof. Starting from equation (*),

$$
\begin{aligned}
&R(f_{G,\hat{\rho}_{\lambda,u}}) \leq R_n(f_{G,\hat{\rho}_{\lambda,u}}) + \frac{1}{\lambda} D_{\mathrm{KL}}(\hat{\rho}_{\lambda,u}, \pi) + U_*(\epsilon; \lambda, n) \\
&= \int R_n(\theta) d\hat{\rho}_{\lambda,u}(\theta) + u \left(\int K_n(\theta) d\hat{\rho}_{\lambda,u} - B_n\left(\hat{\rho}_{\lambda,u}\right) \right) + \frac{1}{\lambda} D_{\mathrm{KL}}(\hat{\rho}_{\lambda,u}, \pi) + U_*(\epsilon; \lambda, n) \\
&= \int R_n(\theta) d\hat{\rho}_{\lambda,u}(\theta) + u \int K_n(\theta) d\hat{\rho}_{\lambda,u} + \frac{1}{\lambda} D_{\mathrm{KL}}(\hat{\rho}_{\lambda,u}, \pi) \\
&\qquad - uB\left(\hat{\rho}_{\lambda,u}\right) + u\left(B\left(\hat{\rho}_{\lambda,u}\right) - B_n\left(\hat{\rho}_{\lambda,u}\right)\right) + U_*(\epsilon; \lambda, n) \\
&\overset{(**)}{\leq} \int R(\theta) d\rho^*_{\lambda,u}(\theta) + u \int K(\theta) d\rho^*_{\lambda,u}(\theta) + \frac{1}{\lambda} D_{\mathrm{KL}}(\rho^*_{\lambda,u}, \pi) \\
&\qquad - uB\left(\hat{\rho}_{\lambda,u}\right) + u\left(B\left(\hat{\rho}_{\lambda,u}\right) - B_n\left(\hat{\rho}_{\lambda,u}\right)\right) + U(\epsilon; \lambda, u, n) \\
&= \underbrace{\int R(\theta) d\rho^*_{\lambda,u}(\theta) + u \left(\int K(\theta) d\rho^*_{\lambda,u}(\theta) - B\left(\hat{\rho}_{\lambda,u}\right) \right) + \frac{1}{\lambda} D_{\mathrm{KL}}(\rho^*_{\lambda,u}, \pi)}_{(1)} \\
&\qquad + \underbrace{u\left(B\left(\hat{\rho}_{\lambda,u}\right) - B_n\left(\hat{\rho}_{\lambda,u}\right)\right)}_{(2)} + U(\epsilon; \lambda, u, n)
\end{aligned}
$$

Idea behind proof of oracle inequality for $u \in \mathcal{W}_u$

For (2), a high probability bound can be derived as in the generalization bound theorem for costs in Section 1.5. For (1), we have

$$\begin{aligned}
&\int R(\theta) d\rho^*_{\lambda,u}(\theta) + u\left(\int K(\theta) d\rho^*_{\lambda,u}(\theta) - B\left(\hat{\rho}_{\lambda,u}\right)\right) + \frac{1}{\lambda} D_{\mathrm{KL}}(\rho^*_{\lambda,u}, \pi) \\
&\leq \sup_{a \geq 0}\left[\int_\Theta R(\theta) d\rho^*_{\lambda,a}(\theta) + a\left(\int_\Theta K(\theta) d\rho^*_{\lambda,a}(\theta) - B\left(\hat{\rho}_{\lambda,u}\right)\right) + \frac{1}{\lambda} D_{\mathrm{KL}}(\rho^*_{\lambda,a}, \pi)\right] \\
&= \min_{\rho \in \mathcal{E}_{B(\hat{\rho}_{\lambda,u})}}\left\{\int_\Theta R(\theta) d\rho(\theta) + \frac{1}{\lambda} D_{\mathrm{KL}}(\rho, \pi)\right\},
\end{aligned}$$

where the last equality has used the most recent lemma.

These arguments lead to the following holding with high probability $(1 - \epsilon)$:

$$R(f_{G,\hat{\rho}_{\lambda,u}}) \leq \min_{\rho \in \mathcal{E}_B(\hat{\rho}_{\lambda,u})}\left\{R(f_{G,\rho}) + \frac{1}{\lambda} D_{\mathrm{KL}}(\rho, \pi)\right\} + \overline{U}(\epsilon; \lambda, n, u).$$

A high probability bound of this form is the main hurdle in the formal proof for the oracle inequality when $u \in \mathcal{W}_u$. Proof for pre-determined B utilizes some similar arguments.

back

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